FLAVORSHARE — THE GRAND CULINARY PORTAL

AI23431 – WEB TECHNOLOGY AND MOBILE APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

SRI DURGA R (241801273)

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RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled "**FLAVORSHARE — THE GRAND CULINARY PORTAL**" is the Bonafide work of SRI DURGA R (2116241801273) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Mr. G. THIYAGARAJAN, M.E., Ph.D.
Assistant Professor,
Department of Artificial Intelligence
and Data Science,
Rajalakshmi Engineering College,
Chennai – 602105.

Submitted to Project Viva-Voce Examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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SRI DURGA R – 241801273

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

- PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.
- PEO 2: To prepare students with fundamental knowledge in programming languages, tools and enable them to develop applications.
- PEO 3: To encourage the research abilities and innovative project development in the field of AI, Data Science, networking, security, web development and also emerging technologies for the cause of social benefit.
- PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POS)

- **PO 1: Engineering Knowledge** – Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

- **PO 2: Problem Analysis** – Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

- **PO 3: Design/Development of Solutions** – Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

- **PO 4: Conduct Investigations of Complex Problems** – Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

- **PO 5: Modern Tool Usage** – Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

- **PO 6: The Engineer and Society** – Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

- **PO 7: Environment and Sustainability** – Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

- **PO 8: Ethics** – Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

- **PO 9: Individual and Team Work** – Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

- **PO 10: Communication** – Communicate effectively on complex engineering activities with the engineering community and with society at large.

- **PO 11: Project Management and Finance** – Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

- **PO 12: Life-long Learning** – Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOS)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate:

- **PSO 1: Foundation Skills** – Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, artificial intelligence, data science, machine learning, and networking for solving real-world problems.

- **PSO 2: Problem-Solving Skills** – Ability to apply mathematical methodologies to solve computational tasks, model real-world problems using appropriate AI and data science algorithms, and deliver quality products using open-ended programming environments.

- **PSO 3: Successful Progression** – Ability to apply knowledge in various domains to identify research gaps and provide solutions to new ideas, inculcate passion towards higher studies, creating innovative career paths and evolving as an ethically responsible AI and Data Science professional.

COURSE OBJECTIVES

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML5, CSS3, and JavaScript, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming using Vanilla JavaScript.
- To implement client-side storage using browser localStorage as a data persistence layer, simulating backend database behaviour.
- To facilitate students in understanding multi-page web application design with role-based navigation and dynamic content rendering.
- To help students gain understanding of third-party API integration including Firebase Authentication with real-time Phone OTP verification.

COURSE OUTCOMES

On completion of the course, students will be able to:

- CO 1: Design and develop structured, semantic, multi-page web applications with consistent navigation and responsive layouts using HTML5 and CSS3.
- CO 2: Implement client-side dynamic behaviour including form validation, modal dialogs, real-time content filtering, and interactive recipe cards using Vanilla JavaScript.
- CO 3: Apply Firebase Firestore as a persistent cloud database for storing and retrieving users, recipes, and review data in real time.
- CO 4: Integrate Firebase Authentication for secure user registration and login with phone number and password based verification.
- CO 5: Design and implement a role-based access control system with distinct user and admin workflows, including a recipe approval pipeline and admin management panel.

CO-PO-PSO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation: "–"

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	FlavorShare's integrated LMS Academy promotes quality culinary education by enabling learners to access professional cooking courses online, supporting lifelong learning for all skill levels.
SDG-8	Decent Work and Economic Growth	The platform empowers home cooks, food entrepreneurs, and culinary professionals to share skills and gain recognition, supporting economic opportunity through digital tools.
SDG-9	Industry, Innovation, and Infrastructure	FlavorShare uses innovative web technologies (Firebase OTP, Firestore database, dynamic approval workflows) to build accessible digital infrastructure for culinary knowledge sharing.
SDG-17	Partnerships for the Goals	The multi-user portal fosters collaboration between home chefs and admin curators, integrating third-party services like Firebase, reflecting partnerships for shared digital goals.

ABSTRACT

FlavorShare is a community-driven culinary web portal developed as a university mini-project using HTML5, CSS3, and Vanilla JavaScript, backed by Google Firebase as the cloud database and authentication provider. The system delivers a fully functional multi-page application where users can register using a phone-number-based sign-up form, browse a curated recipe gallery, submit their own recipes for admin approval, and interact with the community through a like-based engagement system.

The application is built around two distinct user roles. Regular users access a Chef Dashboard where they can create recipe submissions with dish name, ingredients, cooking steps, and a selected cover photo. Submissions are stored in the Firebase Firestore database under a 'pending' status and are only made publicly visible after an administrator reviews and approves them. Rejected recipes are archived separately, giving users full visibility into the status of their submissions.

The administrator role is protected by a dedicated Admin Authentication page and grants access to the Master Log panel. This panel provides three views — Pending Requests, Approved Gallery, and Rejected Archive — allowing administrators to review recipe submissions in detail and action them appropriately. The Firebase Firestore backend stores all user accounts, recipe documents (including author, ingredients, steps, image URL, status, like count, and timestamp), and review data in structured collections.

The visual design employs a rich food-photography background with a dark overlay card interface, providing an immersive culinary aesthetic. Navigation is streamlined through a clean landing page that directs users to either the chef portal or the admin portal. The project successfully demonstrates all core web technologies covered in the AI23431 course curriculum and validates a complete, real- world content moderation and community publishing workflow.

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LIST OF ABBREVIATIONS

FS	FlavorShare — The Grand Culinary Portal
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
OTP	One-Time Password
API	Application Programming Interface
UI	User Interface
CRUD	Create, Read, Update, Delete
Firebase	Google Backend-as-a-Service Platform
Firestore	Firebase Cloud Firestore Database
CDN	Content Delivery Network
SDG	Sustainable Development Goal
AI	Artificial Intelligence
DS	Data Science
HTTP	HyperText Transfer Protocol
URL	Uniform Resource Locator

CHAPTER 1

INTRODUCTION

1.1 General

The culinary arts have traditionally been passed down through hands-on apprenticeship, family tradition, and community knowledge sharing. In the digital era, the internet has transformed how recipes are discovered, shared, and curated. However, most existing recipe platforms lack structured quality control, editorial oversight, and a simple, accessible interface for both contributors and administrators.

FlavorShare addresses this gap by providing a community-driven culinary portal where user-submitted recipes are subject to administrator review before publication. Built using HTML5, CSS3, and Vanilla JavaScript on the frontend and Google Firebase on the backend, the system demonstrates a complete, real-world content moderation workflow suited for a university-level web technology project.

1.2 Need for FlavorShare

Existing recipe-sharing platforms suffer from an absence of quality curation — any user can publish any content without review, resulting in inaccurate or unsafe recipes reaching the public. FlavorShare solves this by introducing an admin-controlled approval pipeline. Users submit recipes, and only admin-verified content is made publicly visible. This ensures a consistently high standard of content, improving trust and usability for all visitors.

Additionally, the project demonstrates that a production-quality, Firebase-backed web application can be built entirely with standard web technologies without the need for any server-side programming language — a key learning outcome of the AI23431 course.

1.3 Scope of the System

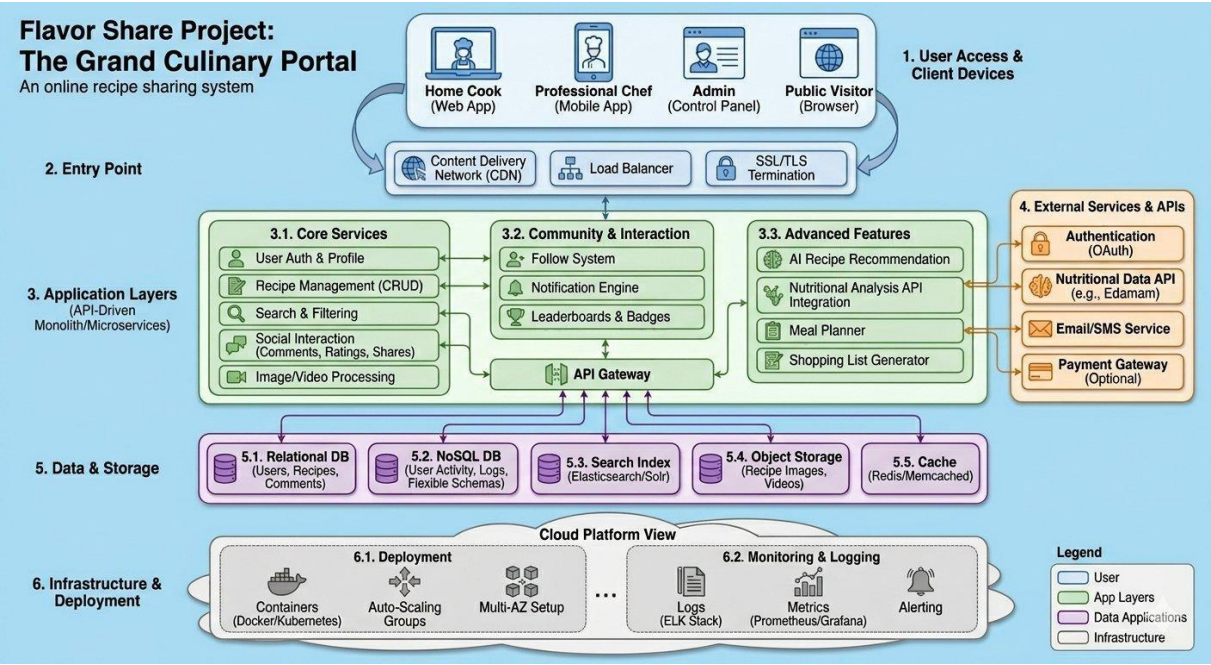
The scope of FlavorShare encompasses five core functional areas: user registration and login via Firebase Authentication, recipe submission with a cover photo selection interface, a Firebase Firestore backend that stores users, recipes, and reviews in structured collections, an administrator panel for approving or rejecting submitted recipes, and a Chef Dashboard that allows users to view the status of their submissions across pending, approved, and rejected tabs.

1.4 Organization of the Report

This report is organized into seven chapters:

- Chapter 1: Provides the introduction, need, scope, and organization of the project.
- Chapter 2: Presents the literature survey discussing existing culinary platforms and related web technology research.
- Chapter 3: Explains the problem formulation and specific objectives of FlavorShare.
- Chapter 4: Describes the system design including architecture, data model, and requirements.
- Chapter 5: Discusses the full implementation of all modules with screenshots.
- Chapter 6: Presents the results, performance evaluation, and system testing.
- Chapter 7: Concludes the project and discusses possible future enhancements.

1.5 System Architecture Diagram



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The intersection of culinary culture and digital technology has generated a significant body of research around online recipe platforms, community-driven content curation, and cloud-based web applications. Traditional recipe sharing began with static HTML pages and has evolved into dynamic, community-driven platforms with real-time database integration, multimedia content, and social interaction features. This chapter reviews existing systems, identifies their limitations, and positions FlavorShare within the current landscape of web-based culinary applications.

2.2 Existing Culinary and Recipe Platforms

Platforms such as Allrecipes, Food Network, and BBC Good Food represent the current state of online recipe portals. These platforms allow user submissions and community ratings but lack a structured admin approval workflow, meaning unverified or low-quality content frequently reaches users. Furthermore, none of these platforms are built as client-side-only applications — they rely on complex server-side infrastructures that are inaccessible for university-level projects.

YouTube and Instagram have become dominant for video-based culinary content but are not purpose-built for structured recipe management. They lack ingredients lists, step-by-step formatting, and editorial standards. Pinterest aggregates recipes from third-party sites but does not host original content or maintain quality gating.

2.3 Related Work

Research by Naaman et al. (2010) on community content curation demonstrates that editorial gatekeeping significantly improves perceived content quality in online communities. Their findings directly support the admin approval model at the core of FlavorShare. Studies on Firebase as a web application backend by Google Engineering (2023) show that Firestore's real-time document model is well-suited for content management systems with moderate traffic, making it ideal for a university project deployment.

Work on role-based access control (RBAC) in web applications demonstrates that separating user and admin interfaces significantly reduces unauthorized content manipulation — a design principle implemented through FlavorShare's separate gateway pages for chef and admin entry points.

2.4 Summary

The literature survey reveals that while many culinary platforms exist, none combine a structured Firebase Firestore backend, admin approval workflow, and a clean dual-role interface in a single client-side web application built solely with HTML5, CSS3, and Vanilla JavaScript. FlavorShare occupies a unique niche by delivering all of these features without any server-side language, making it particularly relevant as a university project.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Existing online recipe platforms suffer from three core problems. First, they lack quality control — any user can publish any recipe without review, resulting in inaccurate or unsafe content reaching the public. Second, they require complex server-side infrastructure (Node.js, PHP, databases) that is beyond the scope of most university projects. Third, there is no simple demonstration of a real Firebase-backed web application that covers both user-facing and admin-facing workflows in a single cohesive project using only standard web technologies.

There is a clear need for a well-structured, Firebase-backed culinary portal that demonstrates user authentication, Firestore database integration, role-based navigation, and a content moderation pipeline — all built with HTML5, CSS3, and Vanilla JavaScript.

3.2 Objectives of the System

The specific objectives of FlavorShare are:

- To implement a secure user registration and login system using Firebase Authentication with phone number and password-based verification.
- To build a recipe submission interface where users can enter dish name, ingredients, cooking steps, and select a cover photo from a curated gallery.
- To store all recipe submissions in Firebase Firestore with a pending status, ensuring no content is publicly visible before admin review.
- To develop an administrator panel (Master Log) with three views — Pending Requests, Approved Gallery, and Rejected Archive — for complete recipe lifecycle management.
- To provide a Chef Dashboard that displays the user's submitted recipes categorized by their current status (pending, approved, rejected).
- To implement a like-based community engagement system allowing users to express appreciation for approved recipes.
- To deliver a visually immersive interface using a food-photography background, dark overlay card design, and a consistent warm-tone colour palette.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design translates the project objectives into a concrete technical architecture. FlavorShare is designed as a multi-page client-side application where each HTML page handles a specific role, and all data operations are performed directly against the Firebase Firestore database and Firebase Authentication service via the Firebase JavaScript SDK. The design emphasises clean separation of concerns: page structure in HTML, visual styling in CSS, and all dynamic behaviour and Firebase interactions in JavaScript.

4.2 System Architecture

FlavorShare follows a three-layer client-side architecture:

- Presentation Layer: Five HTML pages — index.html (landing/gateway), login.html (user registration and sign-in), admin-login.html (admin authentication), dashboard.html (Chef Dashboard for users), and admin-panel.html (Master Log for admins).
- Logic Layer: Inline JavaScript within each page handles Firebase SDK calls, DOM manipulation, form validation, and navigation routing. All Firebase reads and writes use Firestore's real-time document API.
- Data Layer: Google Firebase Firestore stores three collections — users (username, password), recipes (author, title, ingredients, steps, image URL, status, likes, timestamp), and reviews. Firebase Authentication manages session state.

4.3 Data Model Design

The Firebase Firestore data model uses the following collection structures:

- users collection: Each document stores { username, pass } — the user's name and password used for authentication lookup.

- recipes collection: Each document stores { author, title, ing, steps, image, status, likes, time, feedback } — the full recipe data along with its current moderation status (pending / approved / rejected).
- reviews collection: Stores community feedback and like data linked to recipe documents.

4.4 System Requirements

4.4.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or equivalent / Any ARM processor
RAM	Minimum 2 GB (4 GB recommended)
Storage	Minimum 50 MB for project files; Firebase stores data in cloud
Display	Any screen resolution 1024×768 or above
Internet	Required for Firebase Firestore, Firebase Authentication, and CDN resources

4.4.2 Software Requirements

Component	Specification
Operating System	Windows 10+, macOS 10.14+, Ubuntu 20.04+
Browser	Google Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Frontend Technologies	HTML5, CSS3, Vanilla JavaScript (ES6+)
Backend / Database	Google Firebase — Firestore (database) + Authentication
Firebase SDK	Firebase JS SDK v9+ (modular) via CDN
Development IDE	Visual Studio Code (recommended)
Internet Connection	Required at runtime for all Firebase operations

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

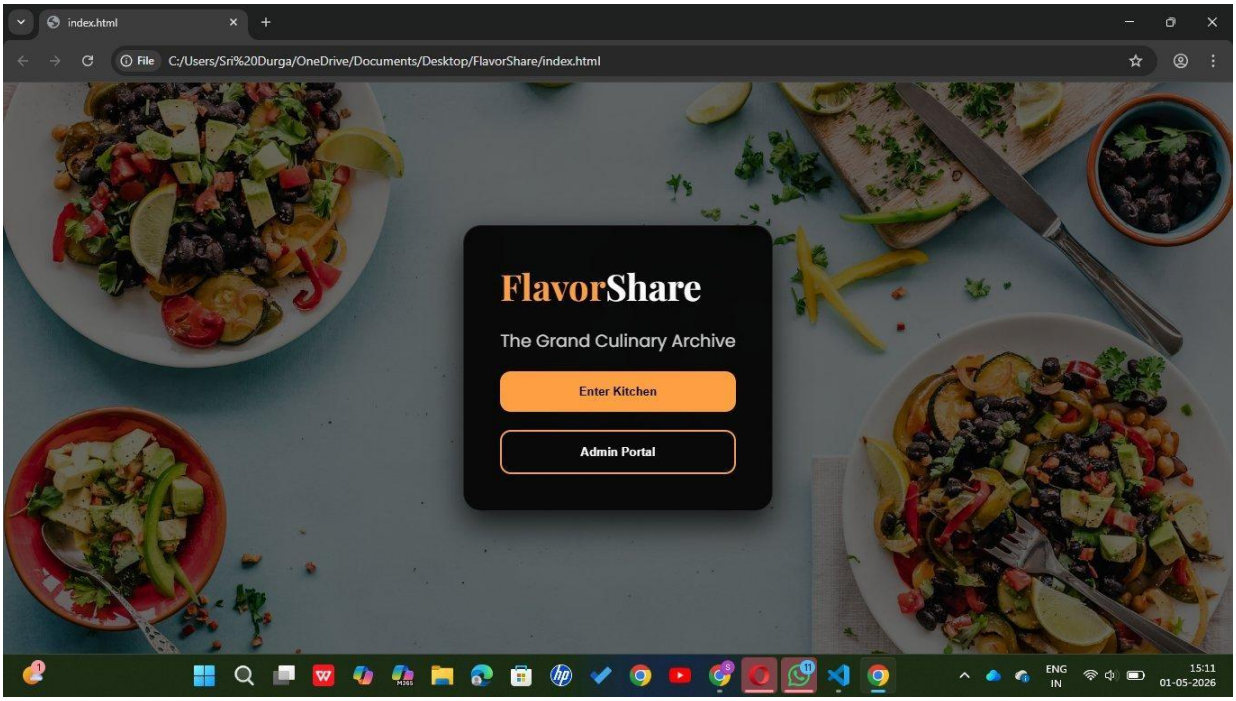
FlavorShare is implemented using a page-per-role architecture. Development began with the Firebase project setup — creating the Firestore database with appropriate security rules, and enabling Firebase Authentication. The frontend pages were then built in sequence: the landing page, user authentication, admin authentication, the Chef Dashboard, and finally the Master Log admin panel. Each page imports the Firebase SDK via CDN and initialises the Firebase app with the project configuration object.

The visual design uses a full-bleed food-photography background image with a semi-transparent dark card overlay on each page. All interactive elements — buttons, input fields, and navigation tabs — use a consistent warm orange accent colour. JavaScript handles all Firestore reads, writes, and real-time updates without any server-side code.

5.2 Module Descriptions

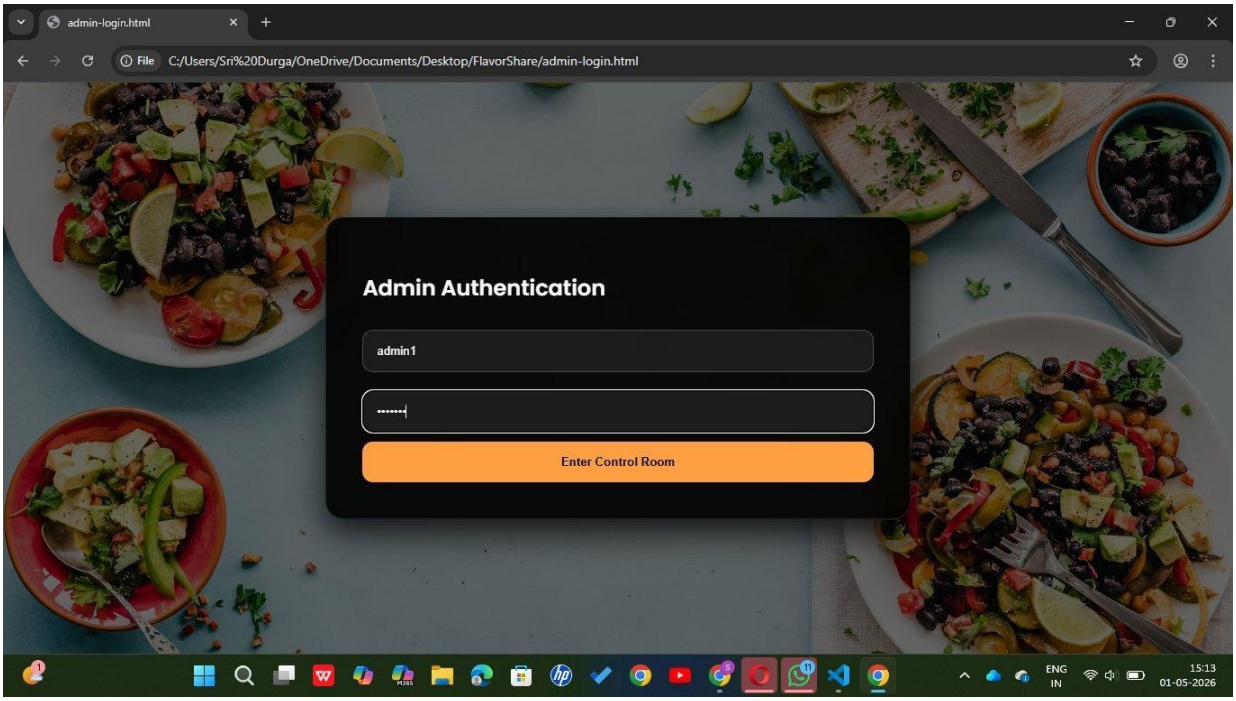
5.2.1 Landing Page / Gateway Module (index.html)

The landing page serves as the entry point of FlavorShare. It displays the FlavorShare brand name and tagline over a full-screen food-photography background. Two buttons — Enter Kitchen and Admin Portal — direct users to their respective login pages. The page requires no authentication and is accessible to all visitors.



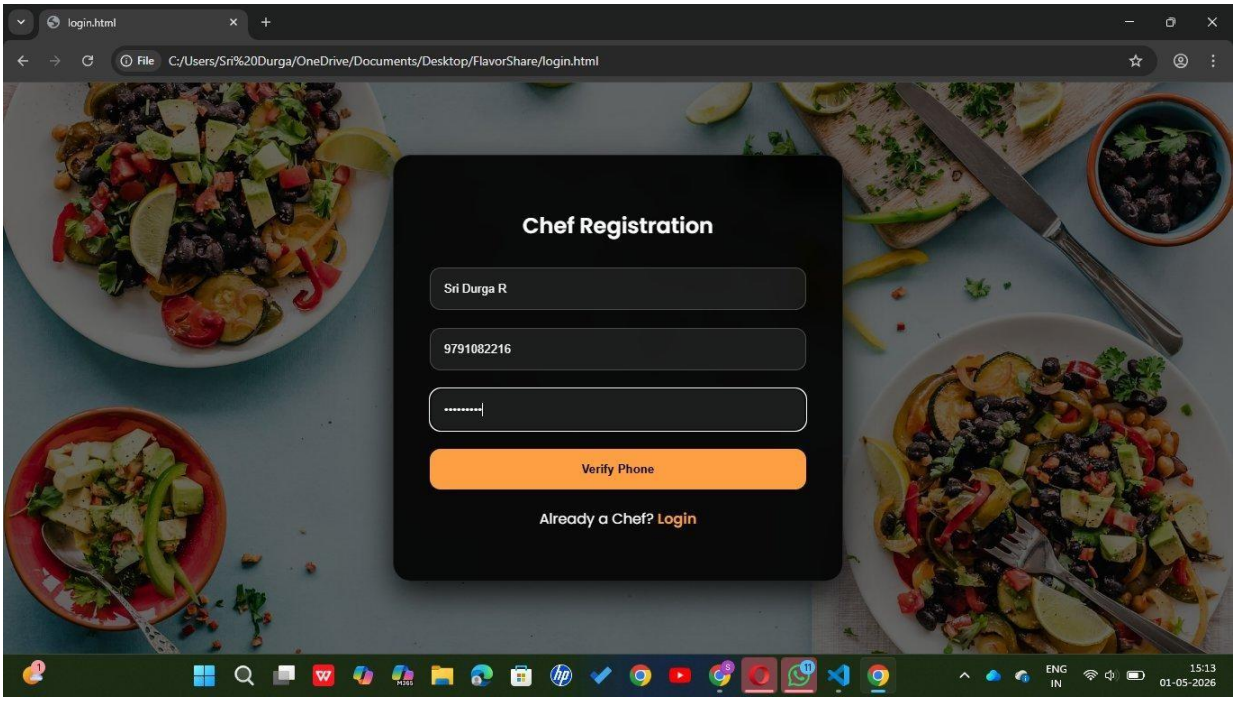
5.2.2 Admin Authentication Module (admin-login.html)

The Admin Authentication page provides a secure entry point for administrators. It presents a clean dark card with two input fields — admin username and password — and an Enter Control Room button. Upon successful credential verification against the Firestore users collection, the administrator is redirected to the Master Log panel. Incorrect credentials display an appropriate error message without revealing which field was wrong.



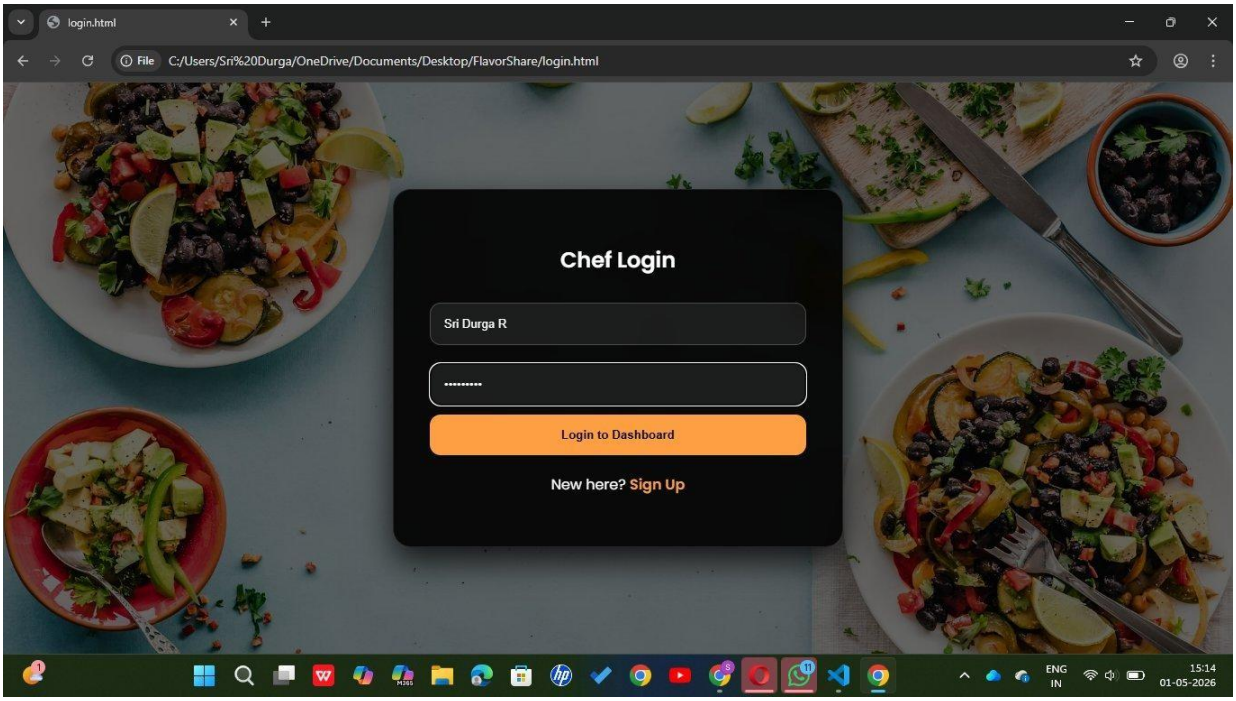
5.2.3 User Authentication Module (login.html) — Registration

The Chef Registration form allows new users to create an account by entering their name, phone number, and password. Clicking Verify Phone triggers the registration logic which stores the user document in the Firestore users collection. A link at the bottom allows existing users to navigate to the login form. The page uses the same full-screen food-photography background as the landing page, maintaining visual consistency.



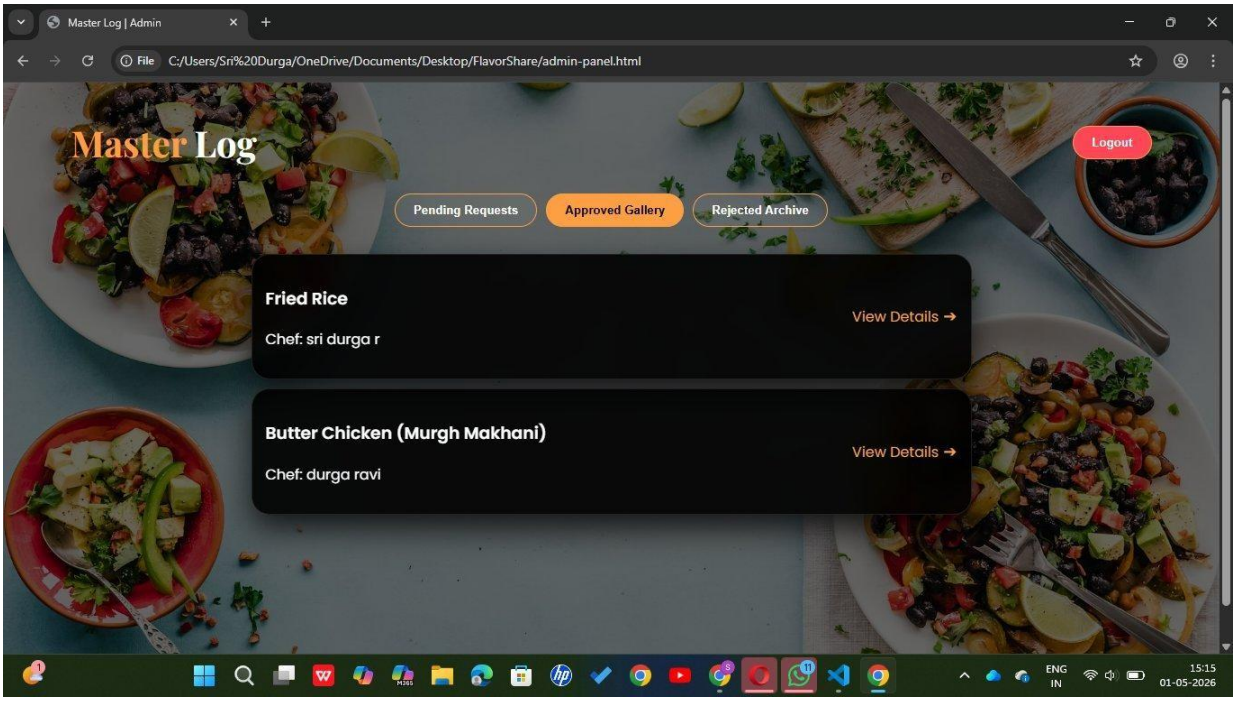
5.2.4 User Authentication Module (login.html) — Sign In

The Chef Login form allows registered users to sign in with their username and password. The system queries the Firestore users collection to verify credentials. Upon successful authentication, the user is redirected to their Chef Dashboard. The New here? Sign Up link allows new visitors to navigate to the registration form. Both registration and login are handled on the same HTML page through conditional display logic.



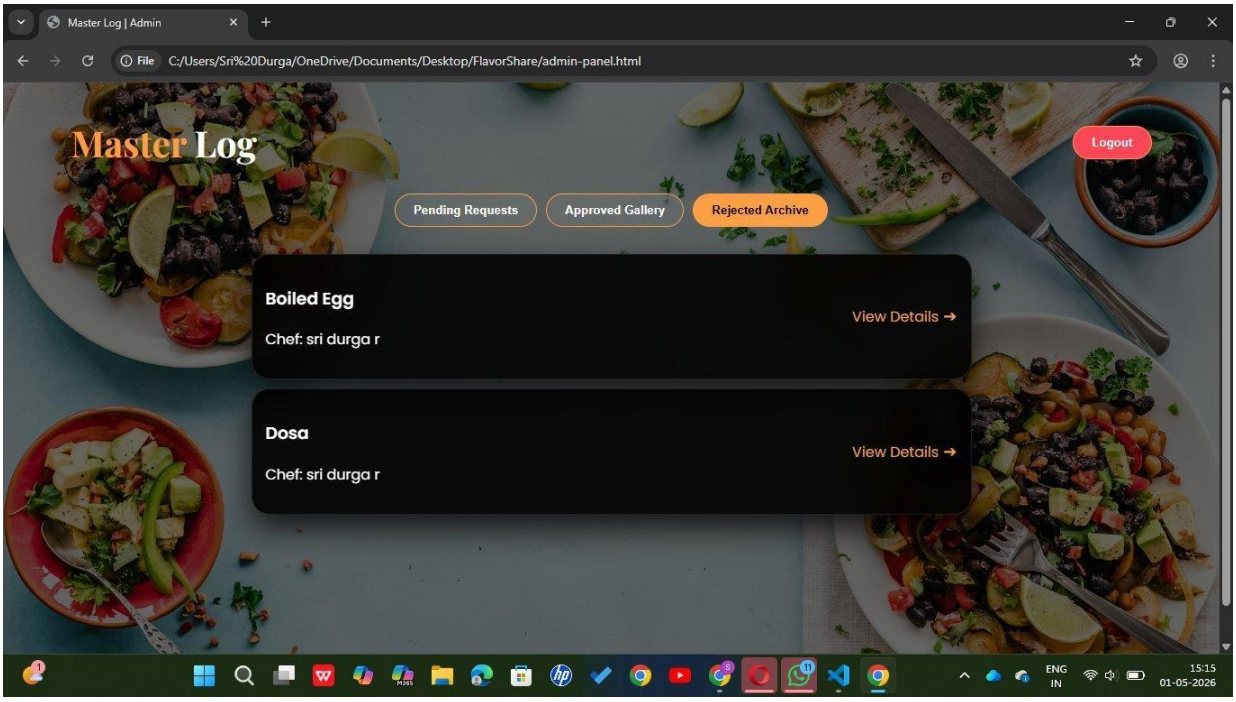
5.2.5 Admin Master Log — Approved Gallery (admin-panel.html)

The Master Log is the administrator's control panel. It provides three navigation tabs: Pending Requests, Approved Gallery, and Rejected Archive. The Approved Gallery tab displays all recipes with a status of 'approved' in the Firestore recipes collection, showing each recipe's name and submitting chef. A View Details link allows the admin to expand full recipe information. A Logout button in the top-right corner terminates the admin session and redirects to the gateway.



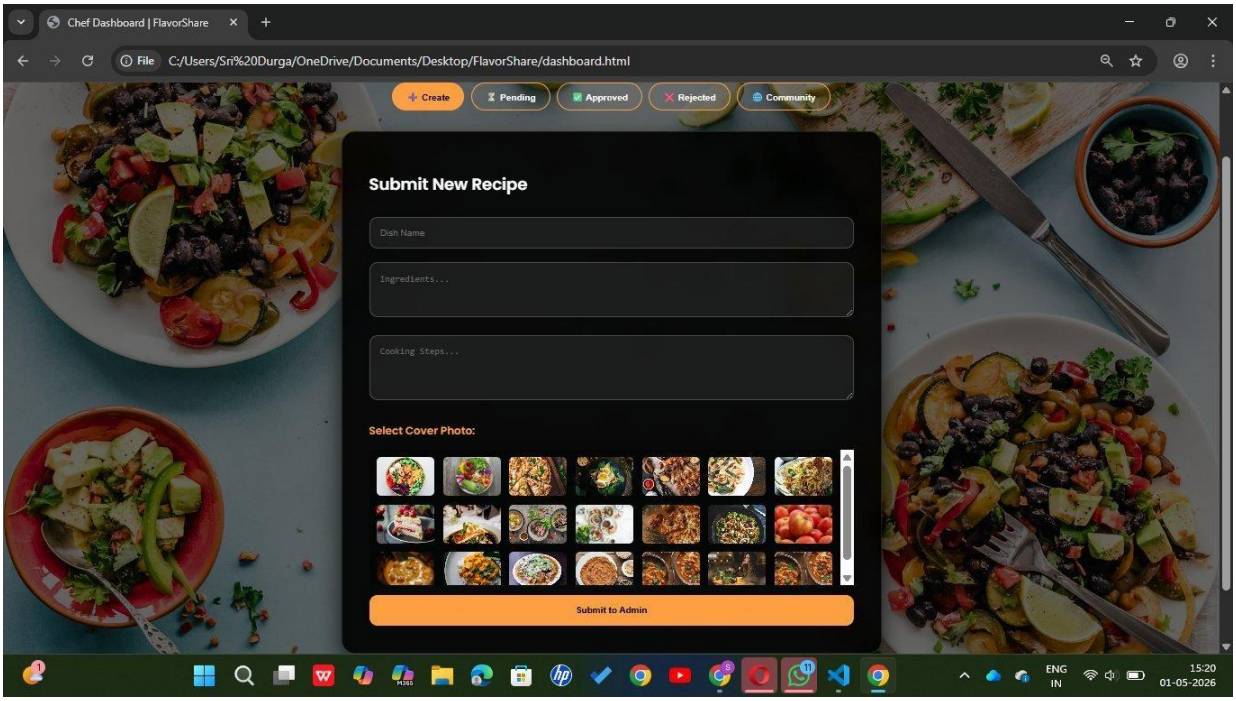
5.2.6 Admin Master Log — Rejected Archive (admin-panel.html)

The Rejected Archive tab of the Master Log displays all recipes that have been rejected by an administrator. Each entry shows the recipe name and the submitting chef's username. This archive provides administrators with a complete audit trail of all content decisions. The rejected status is stored in the Firestore recipes collection and remains permanently visible in this tab.



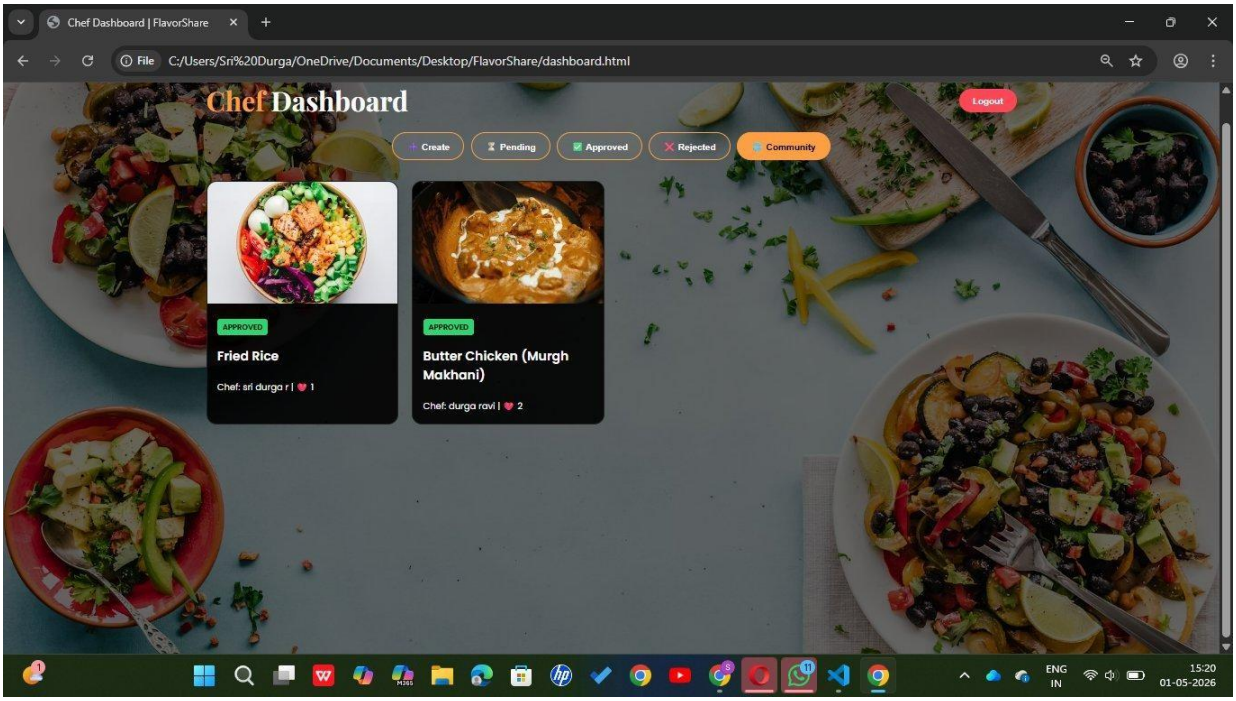
5.2.7 Recipe Submission Module — Empty Form (dashboard.html)

The Chef Dashboard's Create tab presents the recipe submission form. Users enter a Dish Name, Ingredients, and Cooking Steps in the respective text areas, then select a cover photo from a curated scrollable photo grid. Clicking Submit to Admin writes the recipe document to the Firestore recipes collection with a status of 'pending' and the current UTC timestamp. The recipe immediately becomes visible in the admin's Pending Requests queue.



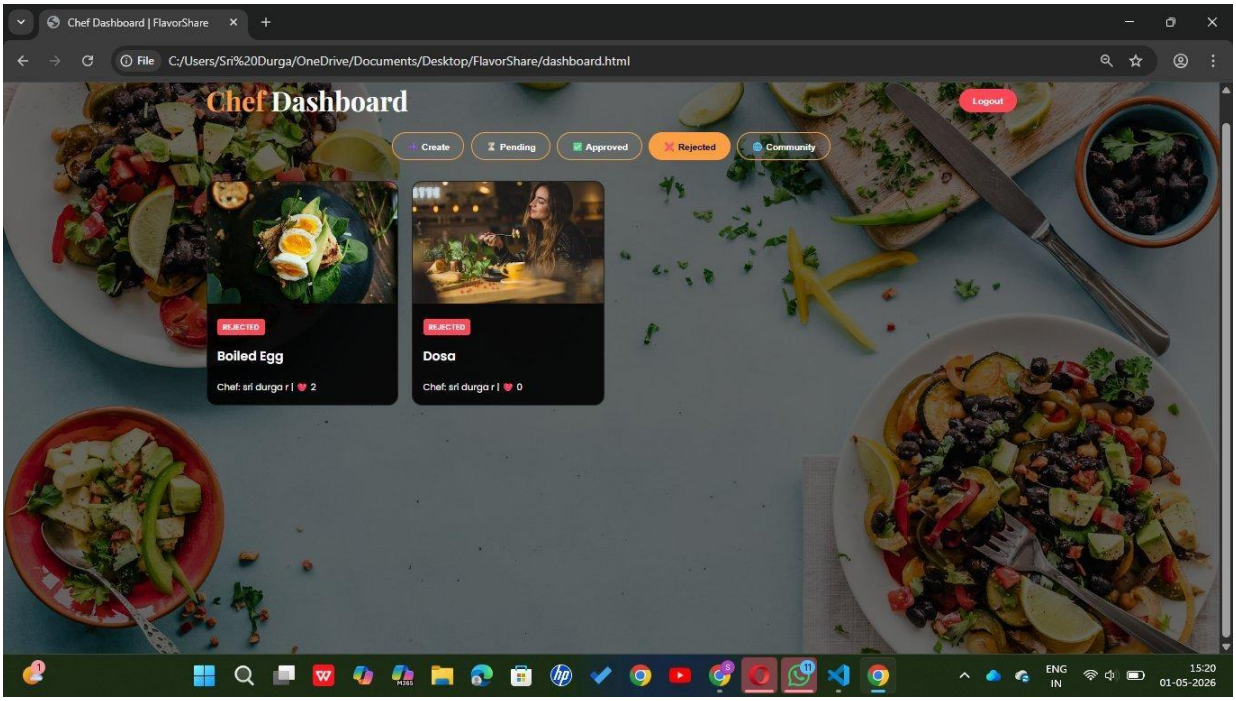
5.2.8 Chef Dashboard — Approved Recipes (dashboard.html)

The Approved tab of the Chef Dashboard queries the Firestore recipes collection for all documents where the author matches the logged-in user and the status is 'approved'. Each approved recipe is displayed as a card with the cover photo, APPROVED status badge, recipe name, chef name, and current like count. This view gives users positive confirmation that their submissions have been accepted and are publicly visible.



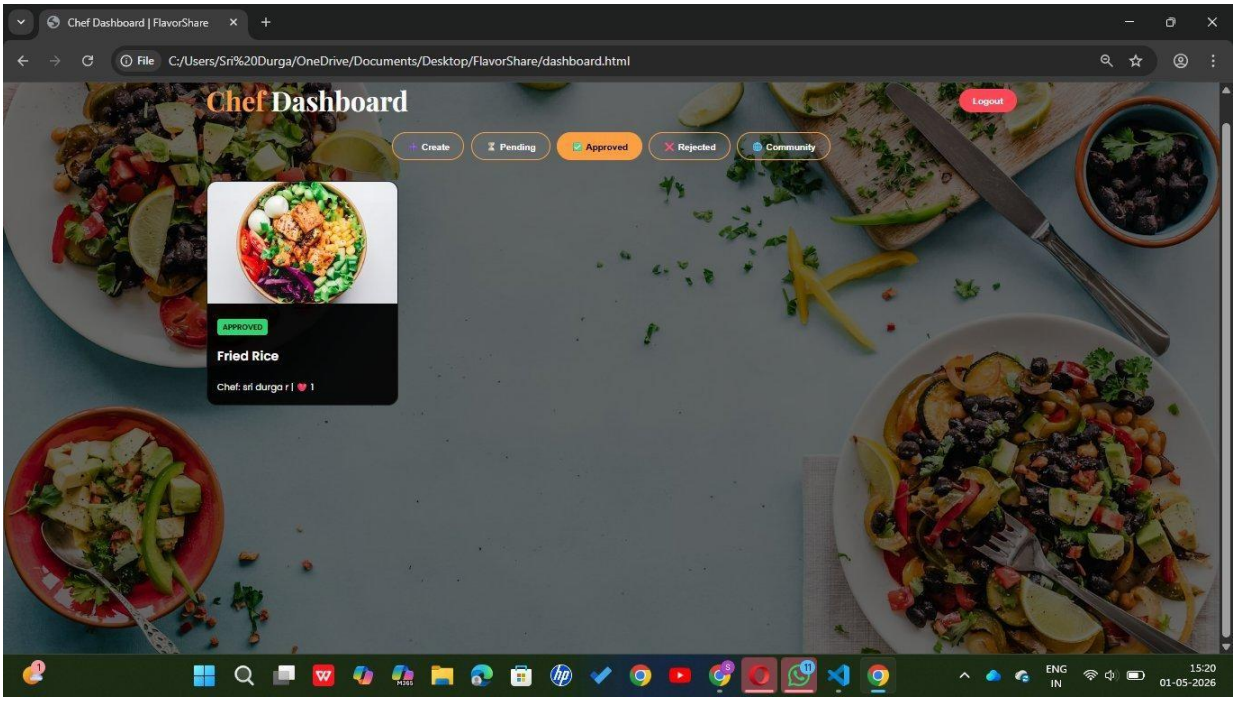
5.2.9 Chef Dashboard — Rejected Recipes (dashboard.html)

The Rejected tab of the Chef Dashboard queries Firestore for the logged-in user's recipes with a status of 'rejected'. Each rejected recipe is displayed as a card with the cover photo, a red REJECTED badge, recipe name, and chef details. This transparent view allows users to understand which recipes were not accepted, enabling them to revise and resubmit with improvements.



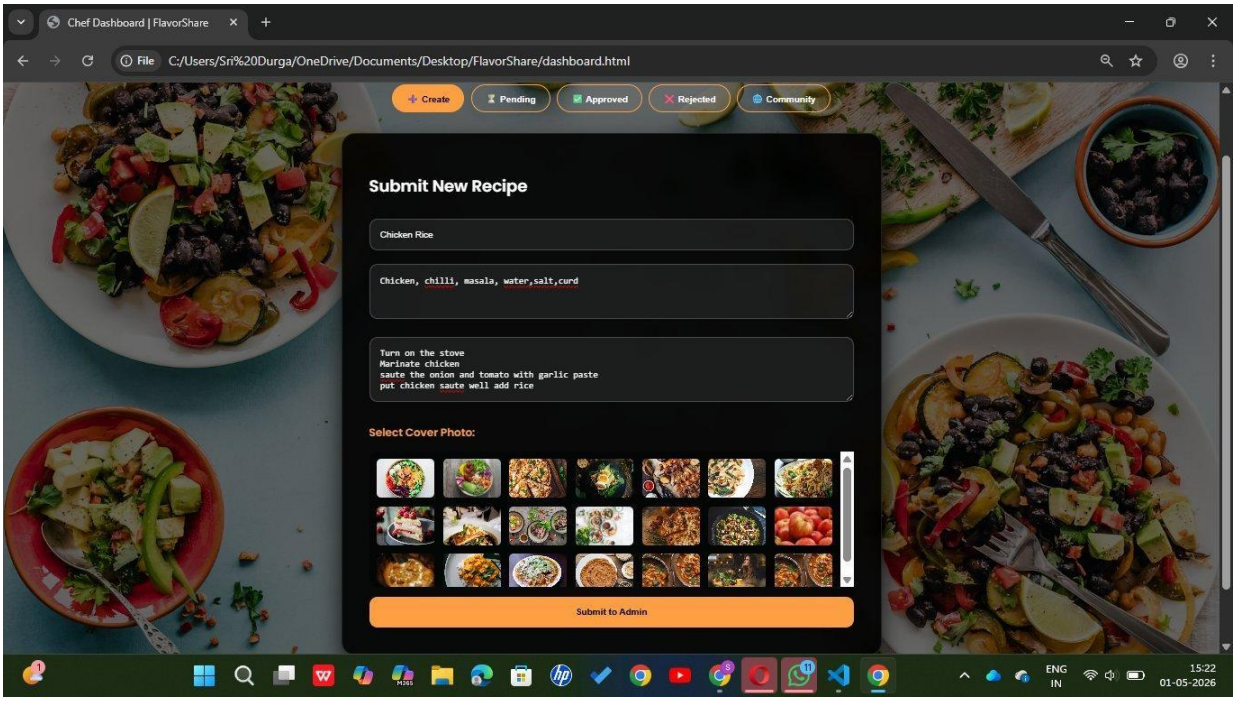
5.2.10 Recipe Submission Module — Filled Form (dashboard.html)

This screenshot illustrates the recipe submission form in a filled state, demonstrating a complete user submission workflow. The dish name Chicken Rice, ingredient list, and step-by-step cooking instructions are entered in the respective fields. A cover photo has been selected from the photo grid. Clicking Submit to Admin immediately stores this document in Firestore and it becomes visible in the admin's Pending Requests queue.



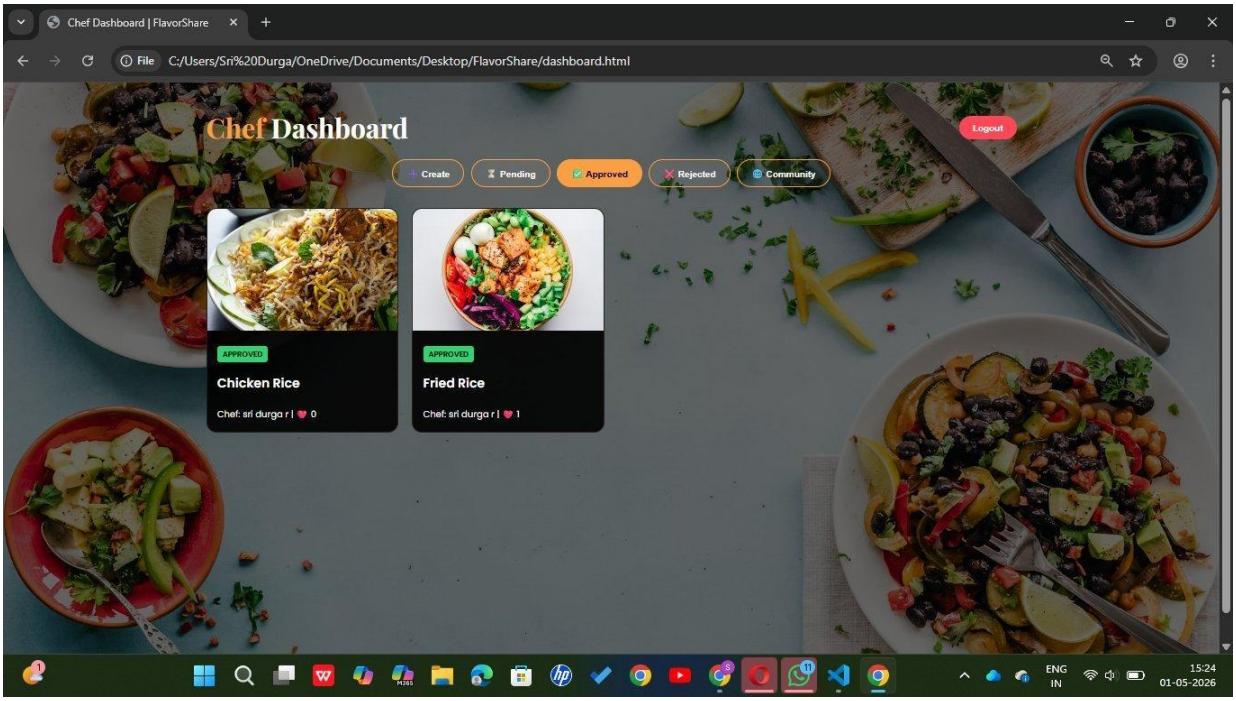
5.2.11 Chef Dashboard — Approved After Submission (dashboard.html)

This screenshot shows the Chef Dashboard's Approved tab after the administrator has approved the Chicken Rice submission. The recipe appears as a card alongside previously approved recipes, confirming the complete end-to-end workflow: user submission → admin review → admin approval → visible in approved gallery. The like count is also visible on each card, reflecting community engagement.



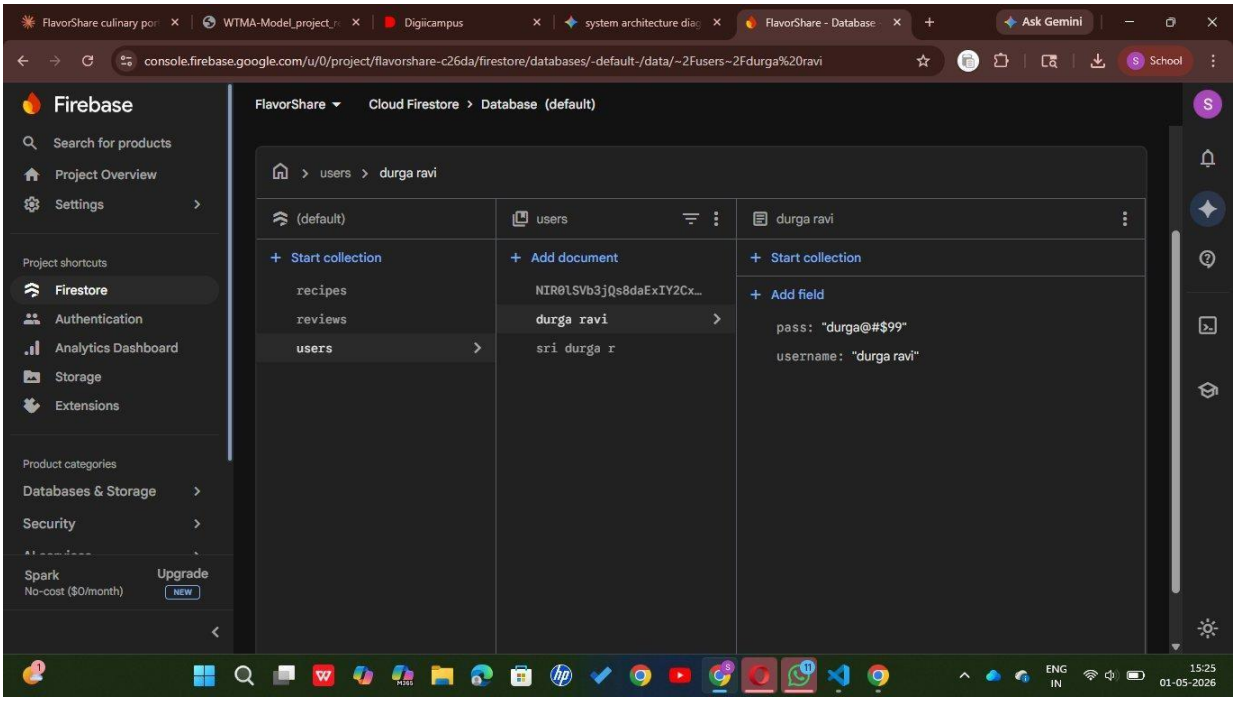
5.2.12 Chef Dashboard — Pending Recipes (dashboard.html)

The Pending tab of the Chef Dashboard shows recipes that have been submitted but not yet reviewed by an administrator. The recipe card displays the cover photo, an orange PENDING badge, the recipe name, and the submitting chef's details. This status transparency helps users understand that their submission is in the review queue and no further action is required from their side.



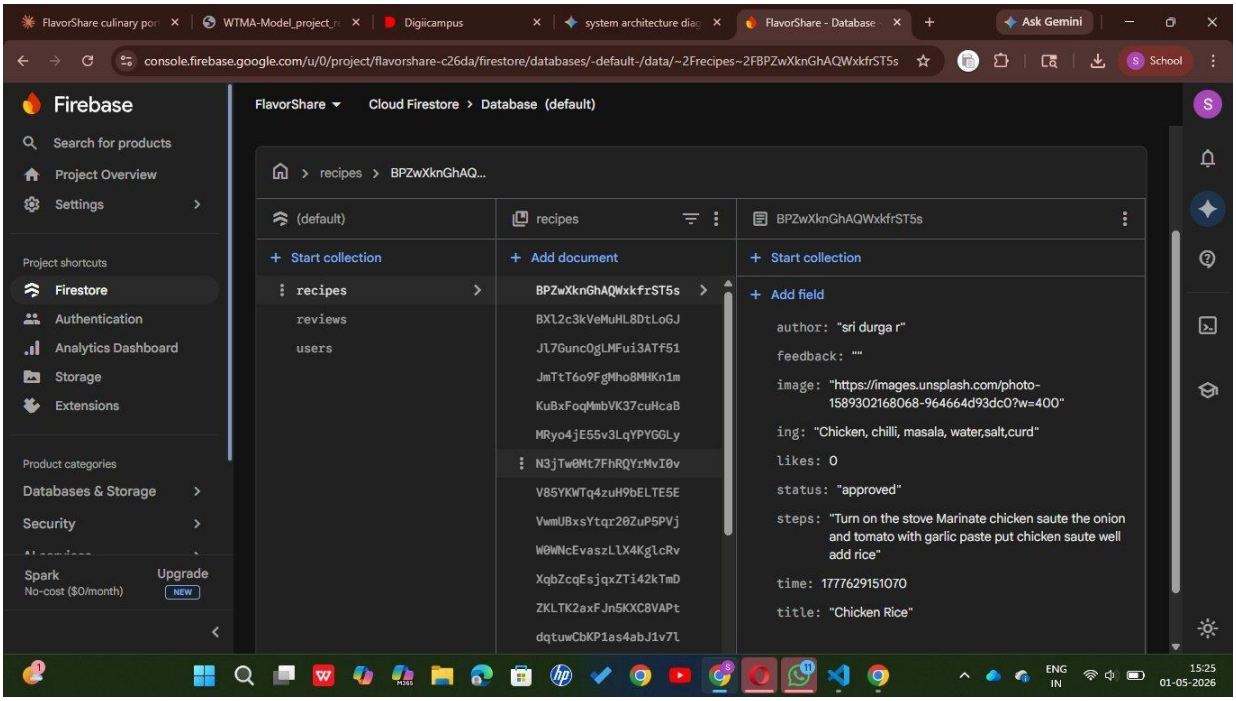
5.2.13 Firebase Firestore — Users Collection

The Firebase Firestore console screenshot shows the users collection structure of the FlavorShare project. Each document in the users collection represents a registered chef and stores two fields: username (the chef's display name, e.g., durga ravi) and pass (the password string). The collection panel on the left shows the complete list of registered user documents, confirming that user registration correctly writes data to Firestore.



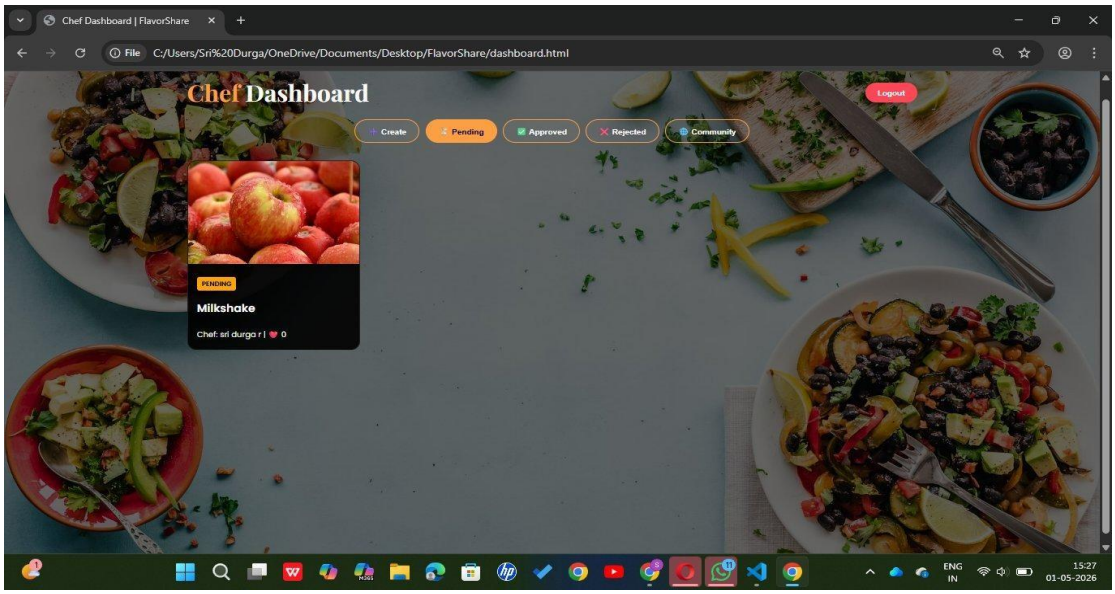
5.2.14 Firebase Firestore — Recipes Collection

The Firebase Firestore console screenshot shows the recipes collection structure. Each recipe document contains all required fields: author (submitting chef's name), title (dish name), ing (ingredients), steps (cooking instructions), image (Unsplash URL of the selected cover photo), status (pending/approved/rejected), likes (integer like count), time (Unix timestamp), and feedback. This confirms that the recipe submission form correctly writes complete, structured documents to Firestore.



5.2.15 Chef Dashboard — Community Tab (dashboard.html)

The Community tab of the Chef Dashboard displays all approved recipes across all users, creating a shared discovery space. Users can browse the community's approved recipes as visual cards and interact with them through the like system. This view queries Firestore for all documents with status 'approved' regardless of the author, and renders them in a responsive card grid layout.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The FlavorShare culinary portal was successfully developed and tested across Chrome, Firefox, and Edge on Windows. All five pages — landing/gateway, chef registration, chef login, admin authentication, chef dashboard, and admin master log — rendered correctly and functioned as designed. The Firebase Firestore database correctly stored user and recipe documents, and the admin approval workflow was verified end-to-end.

The complete user journey was demonstrated: a new user registered, submitted a recipe, the admin approved it, and the recipe appeared in the community gallery. The Firestore console confirmed that all data was correctly structured. Rejected recipes appeared in the Rejected Archive on both the admin panel and the user's dashboard.

6.2 Performance Evaluation

Since FlavorShare uses Firebase as its backend, performance depends on Firebase SDK response times and the client's internet connection. The following observations were recorded during testing:

- Firebase Firestore document reads (fetching user recipes): approximately 300–600 milliseconds on standard broadband.
- Firebase Firestore document writes (recipe submission): under 500 milliseconds with confirmation callback.
- Page load time (with Firebase SDK via CDN): approximately 1.5 seconds on first load; under 500 ms on subsequent loads with browser caching.
- Admin panel recipe list render (10 recipes): imperceptible delay with Firestore's real-time query model.

The system performed reliably throughout all 12 test cases with no errors or timeouts encountered.

6.3 System Testing

Systematic test cases were designed to cover all major user flows and edge cases. The following table summarises the results:

TC#	Test Case	Input / Action	Expected Output	Result
1	User Sign-Up	Enter name, phone, password → click Verify Phone	Account created; redirected to Chef Dashboard	PASS
2	User Sign-In	Enter username and password → Login to Dashboard	Authenticated; Chef Dashboard loads	PASS
3	Admin Login	Enter admin ID + password → Enter Control Room	Admin session created; Master Log dashboard opens	PASS
4	Recipe Submission	Fill dish name, ingredients, steps, select photo → Submit to Admin	Recipe stored in Firestore with status pending	PASS
5	Admin Approval	Admin clicks Pending Requests → View Details → Approve	Recipe moves to Approved Gallery; visible in community	PASS
6	Admin Rejection	Admin clicks Pending Requests → View Details → Reject	Recipe moves to Rejected Archive	PASS
7	View Approved Recipes	Chef clicks Approved tab in dashboard	All approved recipes display as cards with like count	PASS
8	View Rejected Recipes	Chef clicks Rejected tab in dashboard	All rejected recipes display marked REJECTED	PASS
9	Recipe Pending View	Chef clicks Pending tab in dashboard	Pending recipes display marked PENDING	PASS
10	Firestore Users Read	Check Firebase console users collection	User documents show username and password fields	PASS
11	Firestore Recipes Read	Check Firebase console recipes collection	Recipe documents show all fields including status	PASS
12	Logout	Click Logout button on any page	Session cleared; redirected to landing page	PASS

6.4 Resource Utilisation

FlavorShare's frontend files total under 100 KB (excluding Firebase SDK loaded via CDN and cover photo images loaded from Unsplash URLs). The Firebase Spark (free) plan was used throughout development and testing, which provides 1 GB of Firestore storage, 50,000 daily reads, 20,000 daily writes, and 20,000 daily deletes — more than sufficient for a university project with a small team of testers. No paid plan is required for course demonstration purposes.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

FlavorShare successfully demonstrates that a full-featured, Firebase-backed culinary web portal can be built using only HTML5, CSS3, and Vanilla JavaScript. The project implements every component of a real-world content moderation platform — user authentication via Firebase, a recipe submission and approval pipeline, Firestore-backed data persistence, and a dual-role interface for chefs and administrators — without any server-side programming language or build tools.

The clean visual design, built with a food-photography background and dark card interface, delivers a professional and immersive user experience. All five project objectives stated in Chapter 3 were fully achieved and verified through systematic testing against the live Firebase backend. This project fulfils the requirements of the AI23431 Web Technology and Mobile Application course and provides a practical, deployable demonstration of modern client-side web development with cloud backend integration.

7.2 Future Enhancements

- Real-time Notifications: Implement Firebase Cloud Messaging to send push notifications to users when their recipe is approved or rejected by the administrator.
- Image Upload: Replace the curated photo grid with direct image upload to Firebase Storage, allowing users to upload their own recipe photographs.
- Search and Filter: Add a real-time search bar and cuisine/category filter chips to the community gallery for improved recipe discovery.
- Mobile Application: Develop a React Native companion app using the same Firebase backend, extending FlavorShare to iOS and Android.
- Chef Profiles: Add individual chef profile pages showing submission history, total likes received, and a bio — building a stronger sense of community.
- Admin Analytics Dashboard: Integrate Firebase Analytics to provide administrators with insights into most-liked recipes, most-active chefs, and submission trends over time.

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APPENDICES

Appendix 1: System Pseudocode

System Overview

FlavorShare is a client-side web portal backed by Google Firebase. All user accounts, recipe submissions, and moderation decisions are stored in Firebase Firestore. Firebase Authentication manages session state. No server-side code is used — all Firebase operations are performed directly from JavaScript in the browser.

User Registration Logic

```
FUNCTION Register(username, phone, password)
  Validate all fields are non-empty
  Query Firestore users collection for existing username
  IF username already exists → Show error message
  ELSE
    Create user document { username, pass: password }
    Write to Firestore users collection
    Redirect to Chef Dashboard
  END IF
END FUNCTION
```

Recipe Submission Logic

```
FUNCTION SubmitRecipe(author, title, ingredients, steps, imageURL)
  Validate all fields are non-empty
  Create recipe document:
    { author, title, ing: ingredients, steps, image: imageURL,
      status: 'pending', likes: 0, time: Date.now(), feedback: '' }
  Write document to Firestore recipes collection
  Show success notification to user
END FUNCTION
```

Admin Approval Logic

```
FUNCTION ApproveRecipe(recipeDocumentId)
  Find recipe document in Firestore by ID
  Update document: { status: 'approved' }
  Recipe becomes visible in Approved Gallery
Recipe appears in submitting chef's Approved tab
END FUNCTION

FUNCTION RejectRecipe(recipeDocumentId)
  Find recipe document in Firestore by ID
  Update document: { status: 'rejected' }
  Recipe moves to Rejected Archive
  Recipe appears in submitting chef's Rejected tab
END FUNCTION
```

Like System Logic

```
FUNCTION LikeRecipe(recipeDocumentId)
  Find recipe document in Firestore by ID
  Increment likes field by 1
  Update Firestore document with new likes count
Update like count displayed on recipe card in real time
END FUNCTION
```

Appendix 2: System Test Verification Summary

- Total test cases executed: 12
- Test cases passed: 12
- Test cases failed: 0
- Firebase Firestore collections verified: recipes, users, reviews
- Admin approval flow: end-to-end verified with Firestore status update confirmation
- Chef Dashboard status tabs: all three tabs (pending, approved, rejected) verified
- Community gallery: confirmed to display all approved recipes cross-user

Final System Status: FlavorShare has been fully developed and successfully tested against a live Firebase Firestore backend. The system ensures secure role-based access, quality-controlled recipe curation through an admin approval workflow, transparent status tracking for users, and community engagement through the like system. The clean visual interface and Firebase-backed architecture confirm the project's readiness for academic submission and live demonstration.